

Bradley Manzo

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Education:

University of California, Santa Cruz

September 2024 - Current

- MS in Computer Science and Engineering
- Expected Graduation Spring 2026

GPA: 3.71/4.00

University of California, Davis

September 2019 - December 2023

- BS in Computer Engineering, Minor in Technology Management

GPA: 3.43/4.00

Projects:

FPGA-Based CNN Hardware Acceleration

September 2025 - Current

- Developed an embedded AI vision system by integrating an ESP32-C3 with Lattice iCE40 FPGA.
- Targeted hardware through a quantized Pytorch model, utilizing progressive QAT and PACT.
- Designed custom environment to quantize model weights and export to hardware and test bench.
- Constructed a streaming CNN in SystemVerilog parameterized for varying channels and bit widths.
- Achieved 99% LC utilization, sequencing wide layers over DSPs with optimized BRAM buffering.
- Functionally verified using Python Cocotb via unit testing and successively integrated pipeline tests.
- Behaviorally verified FPGA by comparing resulting decision with quantized Pytorch model.
- Explored design space to select globally optimal network, balancing accuracy, latency, and LC cost.

FPGA-Based Tensor Processing Unit

January - December 2023

- Led a group of four in designing a configurable FPGA-based TPU. Built around a systolic array synthesized in Verilog and communicated with through a custom driver program developed in C.
- TPU capable of sparse and dense matrix multiplication up to 4096x4096, 4x faster than software.
- Outfitted with GEMM convolution drivers and ported to Chisel for further parametrization.

Work Experience:

University of California, Santa Cruz, CA

September 2024 – Current

Teaching Assistant

- Guiding students through embedded software development in C using the ESP-IDF framework.
- Teaching protocols such as I2C, SPI, UART, and AXI, as well as IOT deployment and integration.
- Leading daily lab sections, hosting office hours, grading and preparing instruction material.

Tiami Networks, Elk Grove, CA

January - April 2024

FPGA Engineering Intern

- Implemented 5G NR Stack in hardware for SSB Detection using Intel's DSP Builder in Simulink.
- Functionally verified and timed within MATLAB for all valid PSS and SSS sequences.
- Designed and pipelined components such as peak detectors, correlational filters, and accumulators.

Hewlett Packard Enterprise, Roseville, CA

June - September 2023

ASIC-SDK Engineering Intern

- Collaborated with HPE Networking Engineers to identify the cause of degraded network segments.
- Developed internal tests using Python and designed an ML model using the TensorFlow library.
- Designed a deep learning model exhibiting 86% accuracy, effectively diagnosing the degradation.

Skills:

Programming: C/C++, Python, MATLAB, Chisel, Scala, Git, Scripting, Visual Studio Code, UNIX.

Hardware Design: SystemVerilog, Chisel, Cocotb, Quartus, OSS CAD suite, OpenLANE, ESP-IDF.

AI: OCR using Pytesseract, ANN using TensorFlow, CNN using Pytorch, Preprocessing using Pandas.